

Sensitivity-Aware Air Purifier

Innovation dossier - 1. The idea

PROPOSITION

A concept for the real 'Ozone / smell / irritation (weak real signal)' friction, built with the PERSONALISE design move - room/system -> individual/context.

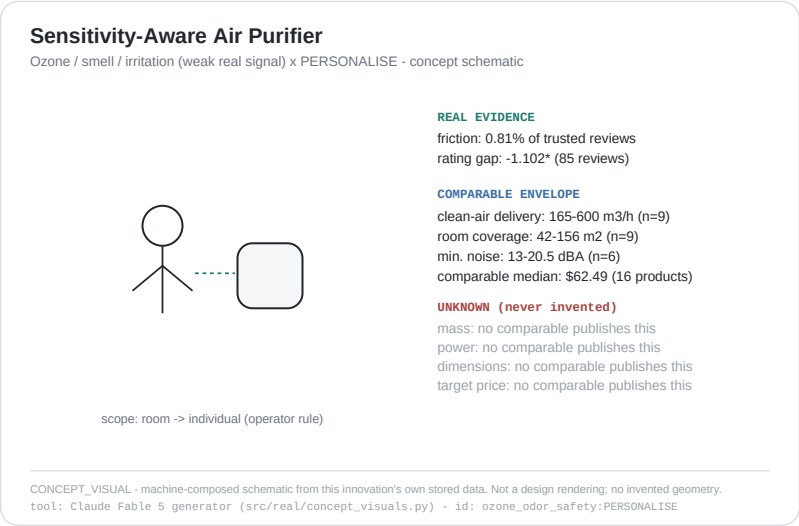


Figure: **CONCEPT_VISUAL**

Machine-composed schematic from the innovation's stored fields only (operator rule, friction stats, comparable envelope). Not a design rendering; a Fable render or drawing would also be CONCEPT_VISUAL, never a digital model or physical prototype.

artifact visual-ozone_odor_safety:PERSONALISE -
/concept-visuals/ozone_odor_safety_PERSONALISE.svg

ENGINEERING ENVELOPE

comparable basis: 10 verified official Versuni/Philips air-purifier families (room-scale)

metric	value from verified comparables	epistemic type
performance cadr m3h	165-600 m3/h (n=9 comparables)	OBSERVED_COMPARABLE
room coverage m2	42-156 m2 (n=9 comparables)	OBSERVED_COMPARABLE
acoustic min dba	13-20.5 dBA (n=6 comparables)	OBSERVED_COMPARABLE
target mass kg	No verified comparable publishes this value - left unknown, never estimated.	UNKNOWN
target power w	No verified comparable publishes this value - left unknown, never estimated.	UNKNOWN
target dimensions	No verified comparable publishes this value - left unknown, never estimated.	UNKNOWN
reference market price usd	median \$62.49 (range \$12.99-\$789.99, n=16) - market price of affected products today, not a proposed price	REFERENCE_MARKET_PRICE

State: **REJECTED**

Killed in the Magic funnel: Another concept beats it on pain, value, and feasibility.

WHY NOW - THE OBSERVED REALITY

0.81% of trusted reviews (85 reviews across 29 real products) carry this friction, with a -1.102* average rating gap. Signal: Ozone/smell/irritation complaints converge with real regulatory guidance on ozone safety limits.

TARGET USER / CONTEXT

Households whose real reviews carry this friction: 85 reviews across 29 real products (82.4% verified purchase, 2011-05-18-2023-04-13).

UNKNOWN - no persona/behavioural dataset exists in this pipeline; only the review evidence above is real.

MECHANISM - METHOD CHOICE

PERSONALISE - Room/system -> individual/context.

METHOD_CHOICE - authored design vocabulary

PRODUCT ARCHETYPE - DESIGN RULE

mobility: Personal/portable - operator PERSONALISE moves scope from room to individual.

room scale, size class, power class, noise ambition, maintenance burden, filter service dependency, connected: UNKNOWN - no real data in this pipeline for this field.

DESIGN_RULE - each populated field follows from the operator's own fixed definition, never from evidence

THE DERIVATION - THREE PARTS, EACH LABELLED

1. Reality **OBSERVED**

0.81% of trusted reviews (85 reviews across 29 real products) carry this friction, with a -1.102* average rating gap. Signal: Ozone/smell/irritation complaints converge with real regulatory guidance on ozone safety limits.

2. Transformation **METHOD CHOICE**

METHOD CHOICE, not evidence: the authored operator table assigns PERSONALISE - "Room/system -> individual/context." - to this friction. Analyst design judgment, applied deterministically.

3. Product consequence **FEASIBILITY PRECEDENT**

Feasibility precedent (from the trend corpus, not a consumer-evidence consequence): Real, directly-relevant regulatory and technical precedent exists: TC-R04 (CARB's real certified-device list is the existing regulatory bar for ozone-safety compliance sensor-led guidance would need to respect) and TC-R08 (Matter's real already-standardized Air Quality Sensor cluster could carry a placement/sensitivity signal without a new protocol).

PARENT PATHS

No parent paths - this innovation cites no tension or assumption paths; its grounding is the friction evidence on page 1.

evidence ids: taxonomy:ozone_odor_safety, TC-R04, TC-R08

DESIGN DNA

F consumer friction (taxonomy:ozone_odor_safety) **PRESENT**

Real Amazon review-text friction theme (see Consumer Pain methodology).

O design operator **METHOD CHOICE**

Authored operator vocabulary applied deterministically (METHOD_CHOICE, not evidence) - see OPERATORS/THEME_OPERATORS in src/real/magic_box_real.py.

S signal (ozone_odor_safety) **PRESENT**

Ozone/smell/irritation complaints converge with real regulatory guidance on ozone safety limits.

T scientific tension **MISSING_UNVERIFIED**

No real research tension shares a paper with this theme's signal evidence.

R rival gap **MISSING_UNVERIFIED**

No real competitor-weakness data clears the white-space threshold for this theme.

C versuni capability **MISSING_UNVERIFIED**

No real Versuni internal capability/org-readiness dataset exists in this pipeline.

A category assumption **MISSING_UNVERIFIED**

No category assumption shares a paper with this theme's signal evidence.

E economic condition **PRESENT**

Real price-weighted exposure computed for this theme (\$16,112.87).

ECONOMICS - RELATIVE INDICATORS ONLY

price-weighted exposure: \$16112.87

SUM of real observed listed prices across affected reviews with a known price. This is a RELATIVE exposure indicator, not a revenue or market-size estimate - it has no units-sold, conversion-rate, or time-period basis. It answers 'which friction touches more expensive products' not 'how much money is at stake per year.'

comparable market median: \$62.49 (n=16 products) - not this concept's price

MEDIAN real listed price across the 16 distinct real products affected by this friction that have a known price. This is what products in this segment actually cost today - not a proposed price for a new concept, which this evidence cannot establish.

category assumptions: No category assumption shares a paper with this theme's signal evidence.

UNCERTAINTIES

- No real Versuni data on current CARB-certification status of the specific product lines this would ship on.

CONTRADICTIONS

- none recorded for this innovation

OBSERVED VS ESTIMATED VS UNKNOWN

observed: friction stats; envelope: performance cadr m3h, room coverage m2, acoustic min dba; DNA F/S/E

method choice: operator, archetype design rule; DNA O

unknown: envelope: target mass kg, target power w, target dimensions; persona; DNA T/R/C/A

Where this idea stands: REJECTED - GRAVEYARD
Killed in the Magic funnel: Another concept beats it on pain, value, and feasibility.

PROTOTYPE STATE

CONCEPT_VISUAL
CONCEPT_VISUAL means a schematic exists, nothing more: no digital model, no physical prototype, no test data.
Machine-composed schematic from the innovation's stored fields only (operator rule, friction stats, comparable envelope). Not a design rendering; a Fable render or drawing would also be CONCEPT_VISUAL, never a digital model or physical prototype.

NEXT EXPERIMENT

Confirmation that the relevant Versuni product lines are already CARB-certified would further strengthen this; the reverse would lower it.

KILL CRITERION

No authored kill criterion yet - a non-formal innovation earns one by displacing the formal recommendation (Scenario lens) or by a human decision to invest.

CRITIC DIMENSIONS

EVIDENCE	CHALLENGE	Gate passed but prevalence (0.81%) is only marginally above the materiality floor - thin evidence base.
ECONOMIC	SURVIVE	Real price-weighted exposure (\$16,113) is at or above the corpus median (\$16,113).
COMPETITIVE	NEEDS_EVIDENCE	No real rival-weakness data recorded for this friction theme.
HUMAN	NEEDS_EVIDENCE	No real usability/human-factors evidence exists in this pipeline for this concept.
PHYSICAL	NEEDS_EVIDENCE	No real lab/engineering feasibility data exists in this pipeline for this concept.
VERSUNI_FIT	NEEDS_EVIDENCE	No real organizational-capability assessment exists in this pipeline for this concept.
TIMING	NEEDS_EVIDENCE	No real market-timing/window evidence exists in this pipeline for this concept.
ROBUSTNESS	NEEDS_EVIDENCE	No real sensitivity/robustness study exists in this pipeline for this concept.
critic overall: CHALLENGE		

CRITERIA RESULTS

NEEDS_EVIDENCE: 35 - PASS: 10 - CHALLENGE: 3 - N/A: 3 - KILL: 1

V8	CHALLENGE	Operator PERSONALISE may add surface area to manage - unverified.
C1	KILL	Not white space.
B2	CHALLENGE	Sensor/automation-dependent theme - see RP-09/RP-10 precision-vs-accuracy tension.
F4	CHALLENGE	ozone_odor_safety theme - real regulatory scrutiny (TC-R02, TC-R04) applies.

LINEAGE / PROVENANCE

generated by: src/real/innovations_real.py
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figure source: /concept-visuals/ozone_odor_safety_PERSONALISE.svg (generated 2026-08-31T16:09:19.527982+00:00 by Claude Fable 5 generator - src/real/concept_visuals.py)
data source: data/processed/innovations_real.json + data/processed/funnel_real.json (deterministic; no invented fields)
dossier generator: src/real/generate_innovation_dossiers.py